

Differential Privacy: definition and main properties



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Differential Privacy

- Increasing volumes of personal and sensitive data are collected and archived by health networks, search engines, social networking systems and other organizations
- A key technological challenge is how to design systems and processing techniques for drawing inferences from these data while maintaining the privacy of the individuals
- Cynthia Dwork and other researchers proposed in 2006 the notion of Differential Privacy



Differential Privacy



Calibrating Noise to Sensitivity in Private Data Analysis

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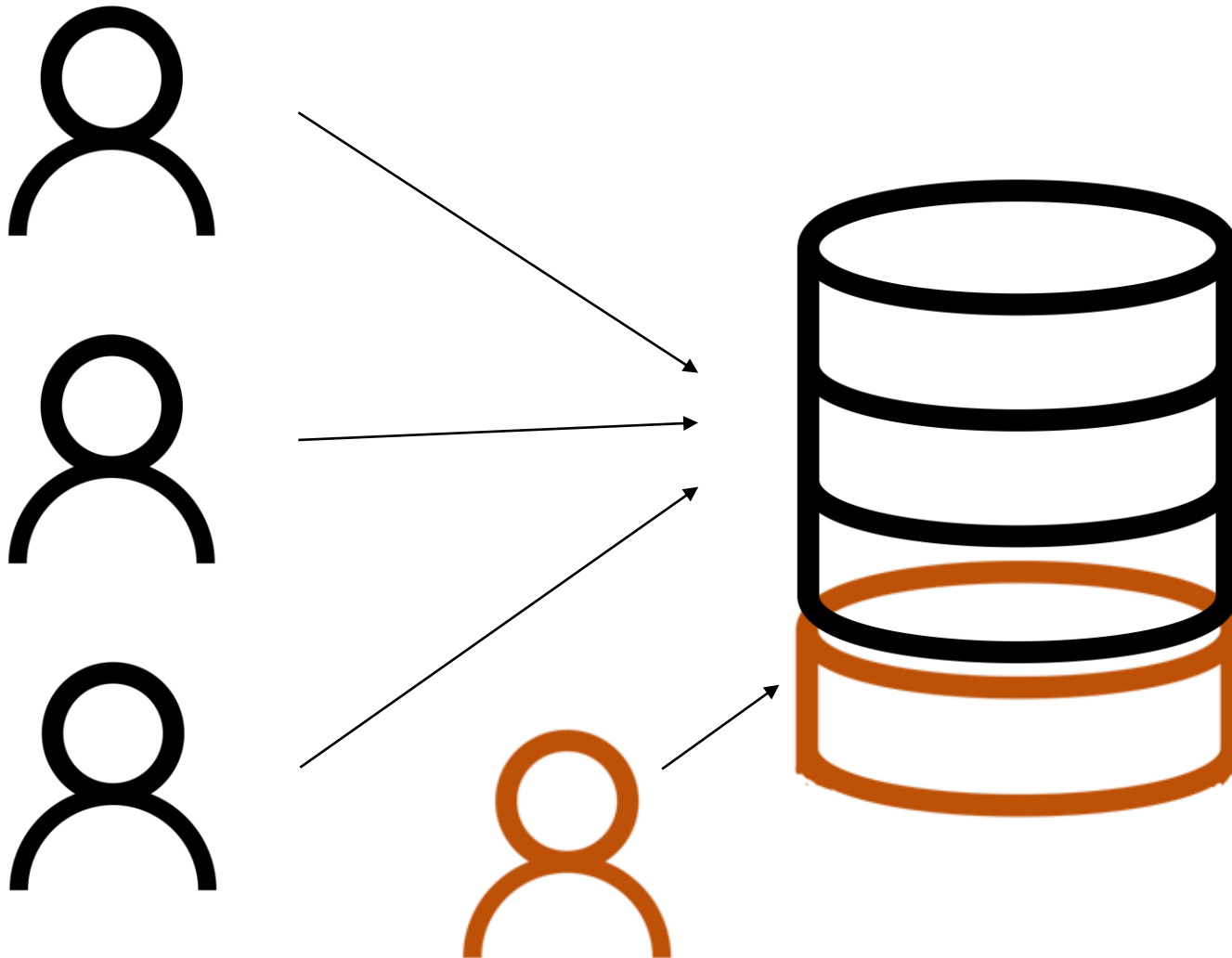
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“Differential Privacy describes a promise, made by a data holder, or curator, to a data subject (owner), and the promise is like this: You will not be affected adversely or otherwise, by allowing your data to be used in any study or analysis, no matter what other studies, datasets or information sources are available.”

C. Dwork

Data owners

Trusted Curator
of the dataset



When we add a record, the result of the query q changes.

For instance,

$$\begin{aligned} q(D) &= 2 \\ q(D \cup \{d\}) &= 2.1 \end{aligned}$$

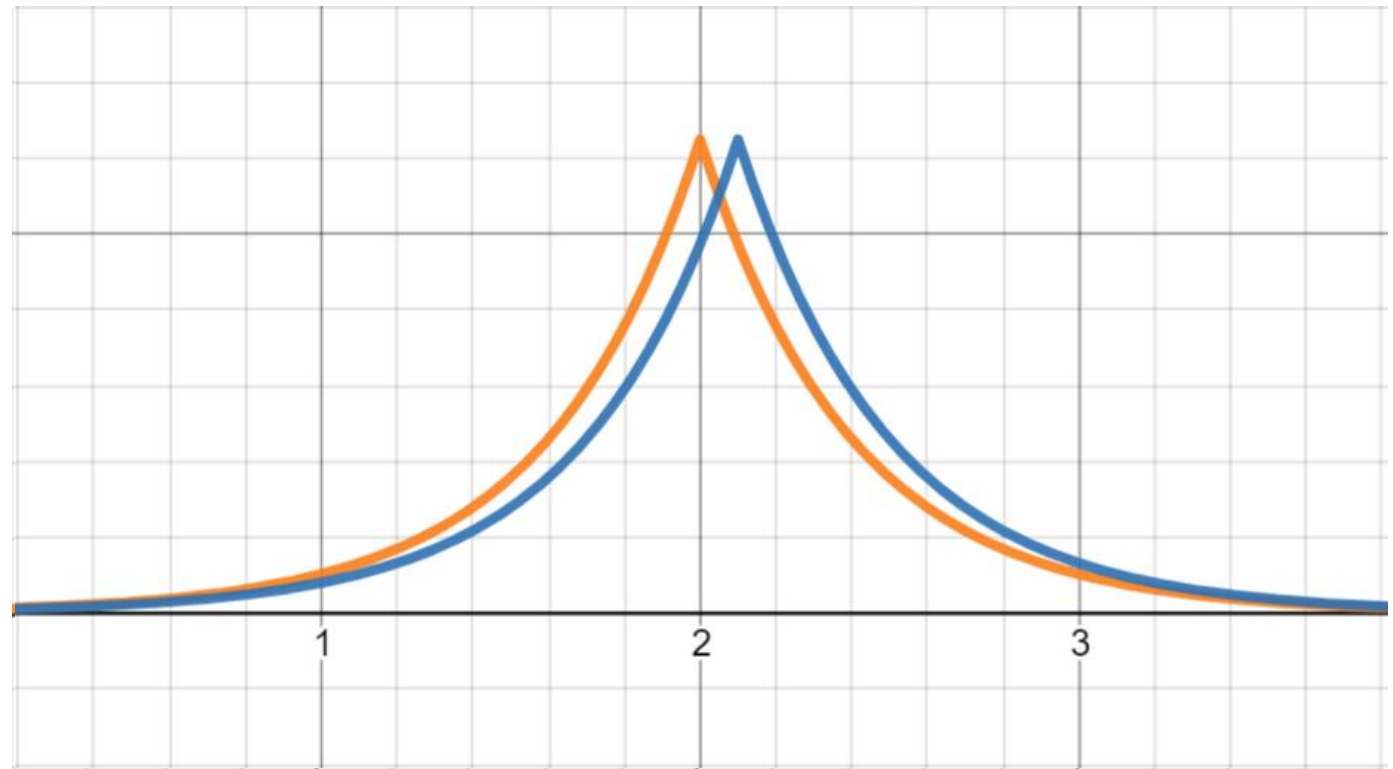
Even when there is only a slight difference between the two values, the result of the query tells us whether the extra record is included in the dataset.

Differential Privacy: idea

$$q(D)=2$$



$$q(D')=2.1$$



Randomized Mechanism

- Publishing the actual result of a query $q: \Omega \rightarrow \mathcal{R}$ may disclose too much information about the secret dataset D .
- The key idea of differential privacy is that some form of randomness must be injected in the computation of $q(D)$, leading to a distorted result $\widetilde{q(D)}$.
- In this way, it is impossible to revert the computation and deterministically understand whether the secret dataset was D or any other similar dataset D' .
- To do that, a *randomized mechanism* is used.

$$\begin{aligned} \mathcal{M}: \Omega \times \mathcal{Q} &\rightarrow \mathcal{R} \\ (D, q) &\mapsto \widetilde{q(D)} \end{aligned}$$

- A randomized mechanism is a stochastic function, i.e. a function that can return any possible value of the codomain $\mathcal{R}$ and each $r \in \mathcal{R}$ has its own probability to be extracted.

ϵ -differential privacy

- Ω : set of all the datasets. $Q = \{q: \Omega \rightarrow \mathcal{R}\}$: set of all the queries.
- Two datasets D, D' are *adjacent* if they differ by only one record: in symbols, $D \sim D'$ if $D' = D \cup \{d\}$ or viceversa.

A mechanism $\mathcal{M}: \Omega \times Q \rightarrow \mathcal{R}$ preserves $\mathcal{E}$ -differential privacy, with $\mathcal{E} > 0$, if, for any query q , any pair of adjacent datasets $D, D' \in \Omega$ and any $r \in \mathcal{R}$,

$$e^{-\epsilon} \leq \frac{P(\mathcal{M}(q, D) = r)}{P(\mathcal{M}(q, D') = r)} \leq e^{\epsilon} \quad \leftarrow \text{Privacy Budget}$$

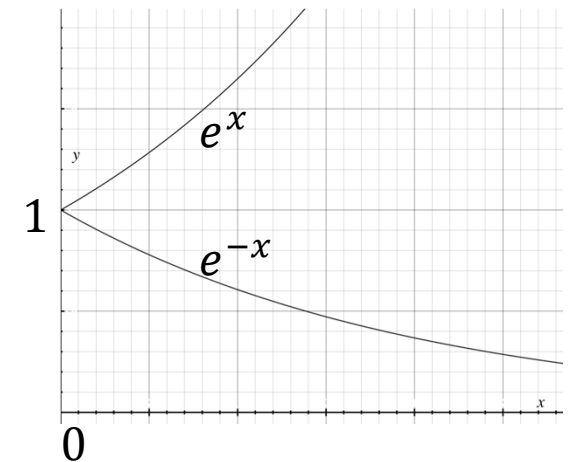
Privacy Budget

- Parameter ε indicates the level of privacy preserved by a differentially private algorithm.
- When $\varepsilon \rightarrow 0$, both $e^\varepsilon, e^{-\varepsilon} \rightarrow 1$. Then, when ε is small,

$$1 \approx e^{-\varepsilon} \leq \frac{P(\mathcal{M}(q, D) = r)}{P(\mathcal{M}(q, D') = r)} \leq e^\varepsilon \approx 1$$

and the condition is equivalent to asking the two probabilities to be almost equal

- The smaller the ε , the stronger the privacy guarantees.



Privacy vs. Accuracy

$$\begin{aligned}\mathcal{M}: \Omega &\rightarrow \mathbb{R} \\ D &\mapsto 100\end{aligned}$$



COMPLETELY
USELESS
MECHANISM!

Mechanism $\mathcal{M}$ perfectly satisfies the differential privacy definition (even with $\varepsilon = 0$): for each pair of adjacent datasets D, D' , we have

$$P(\mathcal{M}(D) = 100) = 1 = P(\mathcal{M}(D') = 100) \cdot e^0$$

$$P(\mathcal{M}(D) \neq 100) = 0 = P(\mathcal{M}(D') \neq 100) \cdot e^0$$

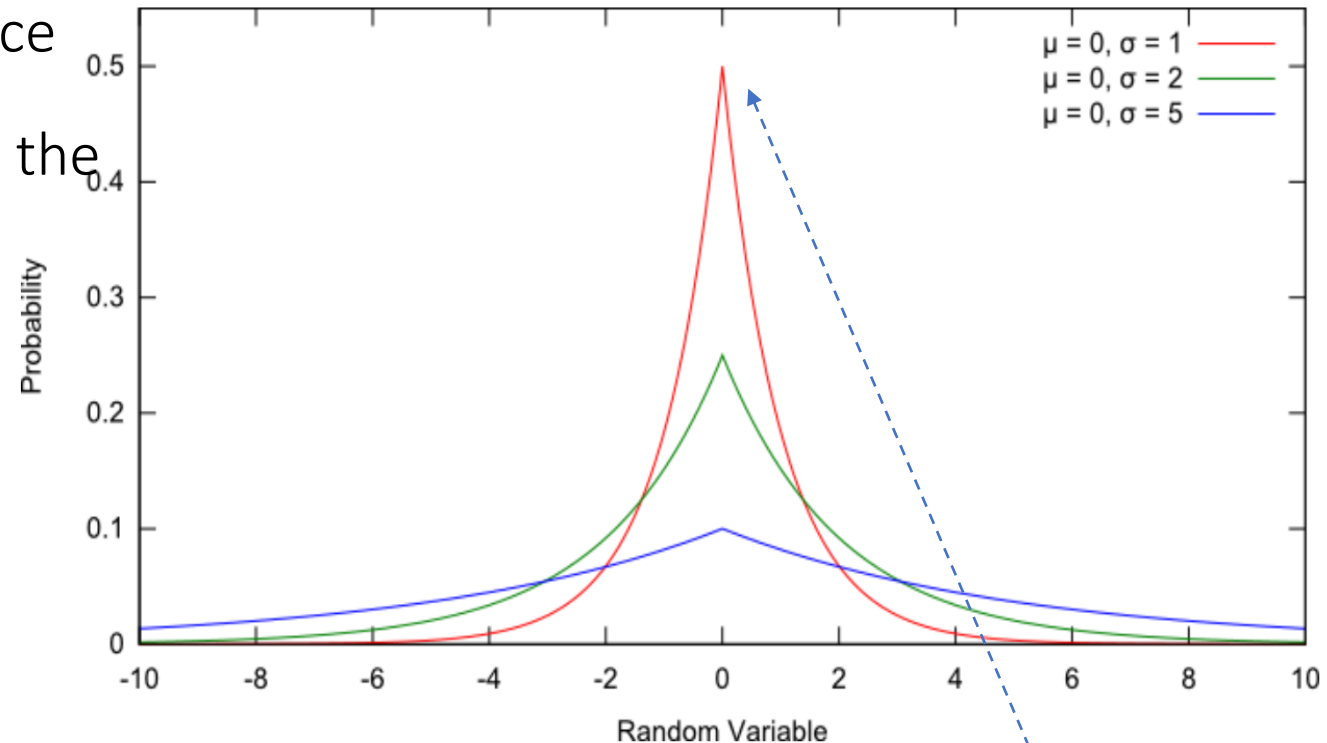
Laplace Mechanism

Given a dataset D and a query q , the Laplace mechanism returns a value drawn from a Laplace distribution $Lap(\mu, \sigma)$ centered in the true result $q(D)$, which in the example has value $q(D)=0$.

$$\mathcal{M}_{Lap}: (D, q) \mapsto Lap(q(D), GS_1(q)/\epsilon)$$

$$\mathcal{M}_{Lap}: (D, q) \mapsto q(D) + Lap(0, GS_1(q)/\epsilon)$$

$$\mathcal{M}_{Lap}: (D, q) \mapsto \frac{\epsilon}{2GS_1(q)} e^{-\frac{\epsilon|x-q(D)|}{GS_1(q)}}$$



$$Lap(x; \mu, \delta) = \frac{1}{2\delta} e^{-\frac{|x-\mu|}{\delta}}$$

Global sensitivity of a function

$GS_1(q)$ is the **GLOBAL SENSITIVITY** of function q , i.e. the measure of the maximum variation of q when computed for two adjacent datasets.

$$GS_1(q) = \max_{D \sim D'} ||q(D) - q(D')||_1$$

The Laplace mechanism with scale parameter $\frac{GS_1(q)}{\epsilon}$ preserves ϵ -differential privacy; on the other hand, if the scale parameter is lower than $\frac{GS_1(q)}{\epsilon}$, DP is no longer guaranteed.

Laplace mechanism is ε -differentially private

Theorem. The Laplace mechanism preserves ε -differential privacy

Proof.

Let $D, D' \in \Omega$ be two adjacent datasets and $q(\cdot)$ some query $q: \Omega \rightarrow \mathcal{R}$.

Let p_D denote the probability density function of $\mathcal{M}_{Lap}(q, D)$ and $p_{D'}$ denote the probability density function of $\mathcal{M}_{Lap}(q, D')$. For an arbitrary result $r \in \mathbb{R}$,

$$\frac{p_D(r)}{p_{D'}(r)} = \frac{e^{-\frac{\varepsilon|q(D)-r|}{GS_1(q)}}}{e^{-\frac{\varepsilon|q(D')-r|}{GS_1(q)}}} = e^{\frac{\varepsilon(|q(D')-r|-|q(D)-r|)}{GS_1(q)}}$$

Laplace mechanism is ϵ -differentially private

$$\frac{p_D(r)}{p_{D'}(r)} = \frac{e^{-\frac{\epsilon|q(D)-r|}{GS_1(q)}}}{e^{-\frac{\epsilon|q(D')-r|}{GS_1(q)}}} = e^{\frac{\epsilon(|q(D')-r|-|q(D)-r|)}{GS_1(q)}}$$

From the triangle inequality, it follows that

$$\frac{p_D(r)}{p_{D'}(r)} = e^{\frac{\epsilon(|q(D')-r|-|q(D)-r|)}{GS_1(q)}} \leq e^{\frac{\epsilon(|q(D')-q(D)|)}{GS_1(q)}} = e^{\frac{\epsilon(\|q(D')-q(D)\|_1)}{GS_1(q)}}$$

But, since $GS_1(q) = \max_{D \sim D'} \|q(D) - q(D')\|_1$ and D, D' are two adjacent datasets, then

$$\frac{p_D(r)}{p_{D'}(r)} \leq e^{\frac{\epsilon(\|q(D')-q(D)\|_1)}{GS_1(q)}} \leq e^\epsilon$$

Laplace Mechanism

In the previous example, the query q to be computed preserving DP returns a single numeric value

$$q: \Omega \rightarrow \mathbb{R}$$



$$\mathcal{M}_{Lap}(q, D) = q(D) + X$$

Laplace Noise



The Laplace mechanism can be used also when q returns a numeric vector

$$q: \Omega \rightarrow \mathbb{R}^n$$



$$\mathcal{M}_{Lap}(q, D) = q(D) + (X_1, \dots, X_n)$$

Laplace Noise
on each component

Example – Counting queries (1)

Dataset D: attendance register today.

We want to compute n (the number of students present) in a DP way, using the Laplace mechanism.

- Privacy Budget: fix $\epsilon = 1$.
- Global sensitivity: adding/removing a record (i.e. adding/removing a student from the class) the total number of students only varies by 1. Then $GS_1 = 1$.

$$\tilde{n} = n + Lap(0,1)$$

Example – Counting queries (2)

Dataset D: attendance register today.

Query $q(D) = (n, n_c)$, where n is the total number of students attending the lesson and n_c is the number of students physically in class.

- Privacy Budget: fix $\varepsilon = 1$.
- Global sensitivity: $GS_1(q) = \max_{D \sim D'} ||q(D) - q(D')||_1 = \max_{D \sim D'} (|n - n'| + |n_c - n'_c|)$
 - Case 1: the extra student is in class: $(|n - n'| + |n_c - n'_c|) = |n - (n + 1)| + |n_c - (n_c + 1)| = 2$
 - Case 2: the extra student is online: $(|n - n'| + |n_c - n'_c|) = |n - (n + 1)| + 0 = 1$
 - $GS_1(q) = 2$

$$GS_1(q) / \varepsilon$$

$$(\tilde{n}, \tilde{n}_c) = (n + \text{Lap}(0, 2), n_c + \text{Lap}(0, 2))$$

Beyond the Laplace Mechanism

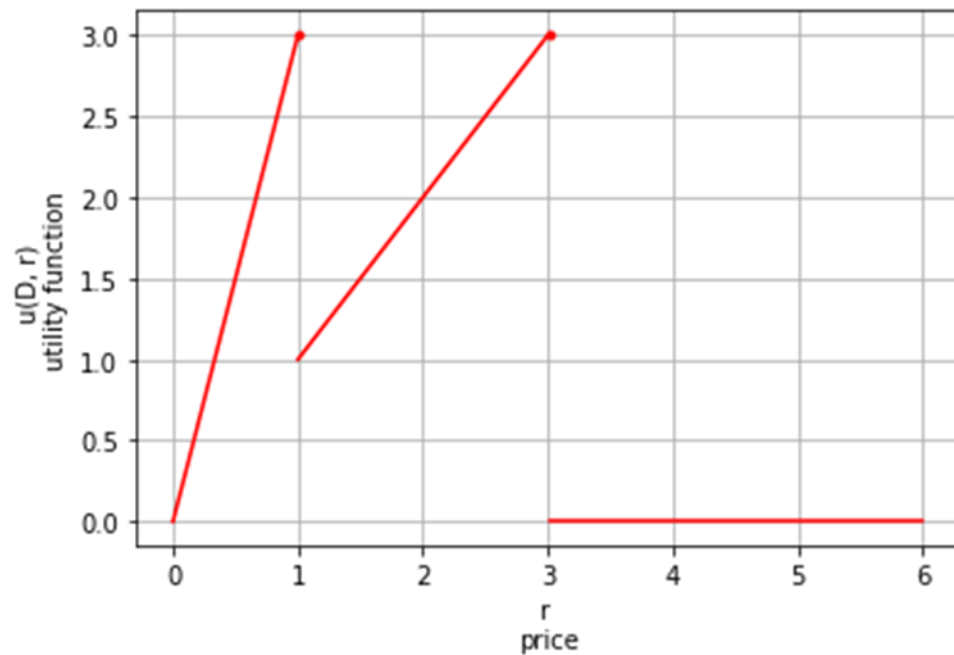
There are cases when the Laplace mechanism cannot be used or does not give an accurate response.

- Non-numeric queries: Laplace mechanism adds numeric noise to the result $q(D)$, but this does not make any sense when $q(D)$ is not a number!
- Differentially private selection: imagine that q returns, among all the possible results, the one that maximizes a utility function U . An additive mechanism (as Laplace mechanism) may destroy the result of q , because even a little noise added to $q(D)$ could correspond to a big change in the utility function.

Differentially private selection - Example

- Dataset D shows how much each customer is willing to pay for a item we want to sell.
- We want to select the optimal prize, the one that (based on how much each possible client is willing to pay) maximises our revenue.
- The optimal prize is 3.01, as shown in the plot below.

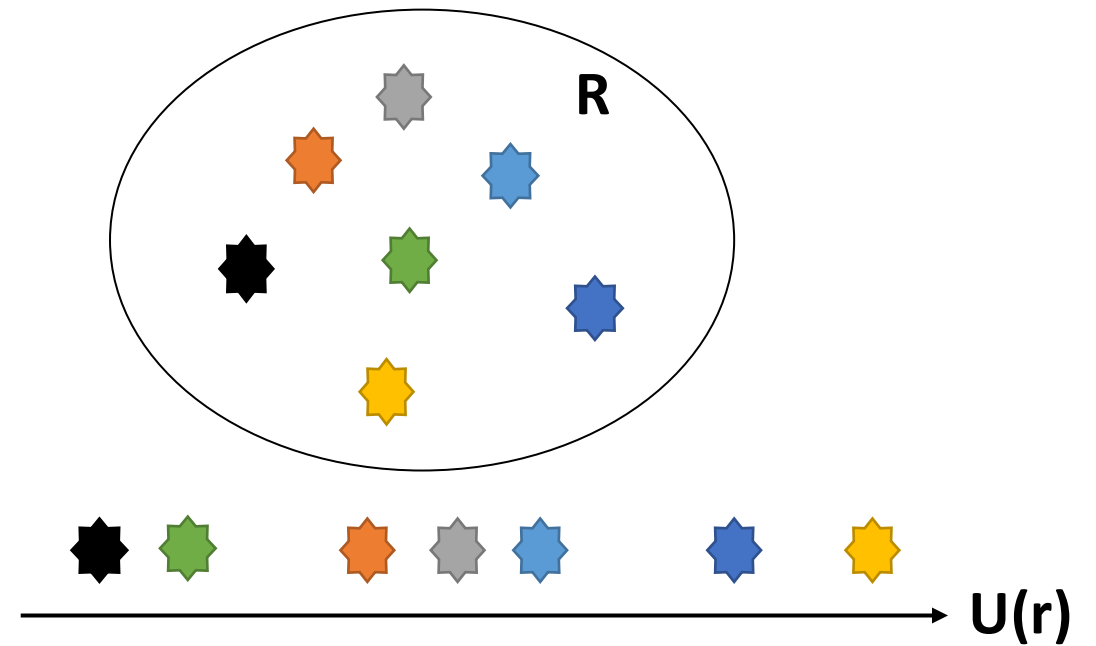
Customer	Max Amount
A	1
B	1
C	3.01



- Let us suppose to use the Laplace mechanism. Instead of publishing the correct prize 3.01, the mechanism adds a small amount of noise and returns 3.02.
- While the variation is very small (only 0.01), the revenue – i.e. the utility function we want to maximize – falls down to 0.

Exponential Mechanism

- Let R be the set of possible outcomes of a query q
- q returns the value $r \in R$ which maximizes a utility function $U: \Omega \times R \rightarrow \mathbb{R}$
- The exponential mechanism assigns to each $r \in R$ a probability value $p(r)$ proportional to its utility $U(r)$



Exponential Mechanism

The exact shape of the probability distribution depends on two factors:

- The privacy budget ϵ
- The utility function U we want to maximize

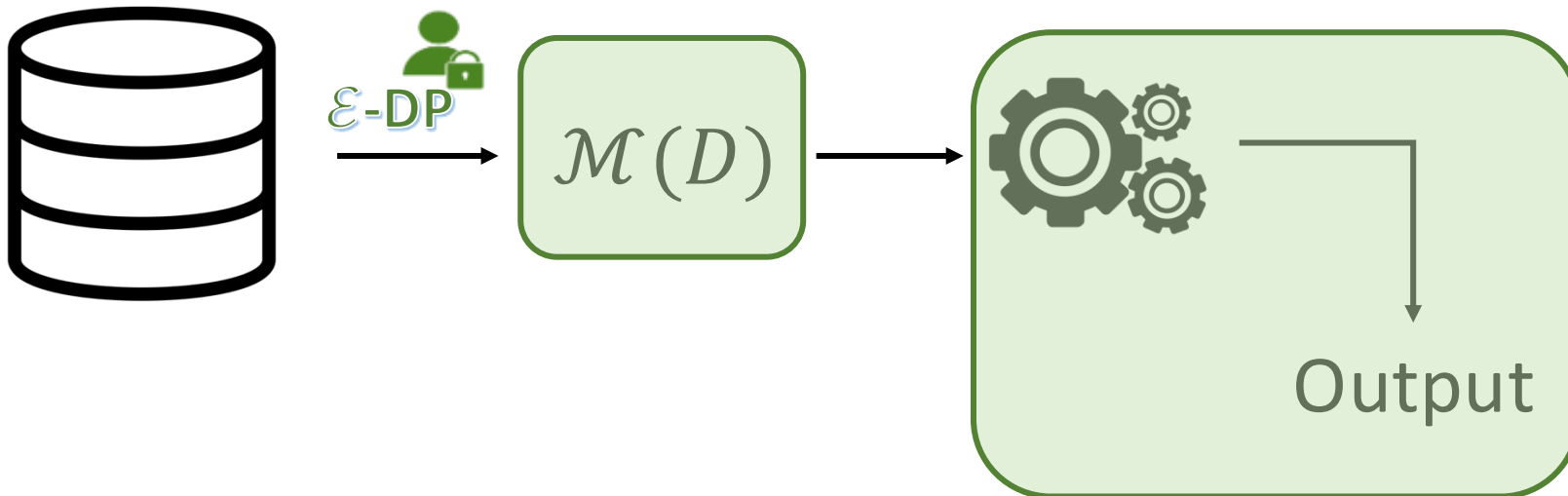
$$p(r) \propto e^{\frac{\epsilon U(D,r)}{2 GS(U)}}, \text{ where}$$

$$GS(U) = \max_{r \in R} \max_{D \sim D'} |U(D, r) - U(D', r)|$$

Differential Privacy properties

Post-Processing Theorem

Let $\mathcal{M}: \Omega \rightarrow \mathcal{R}$ be a mechanism preserving ϵ -differential privacy. Let f be any function with domain $\mathcal{R}$. Then $f \circ \mathcal{M}$ preserves ϵ -differential privacy.

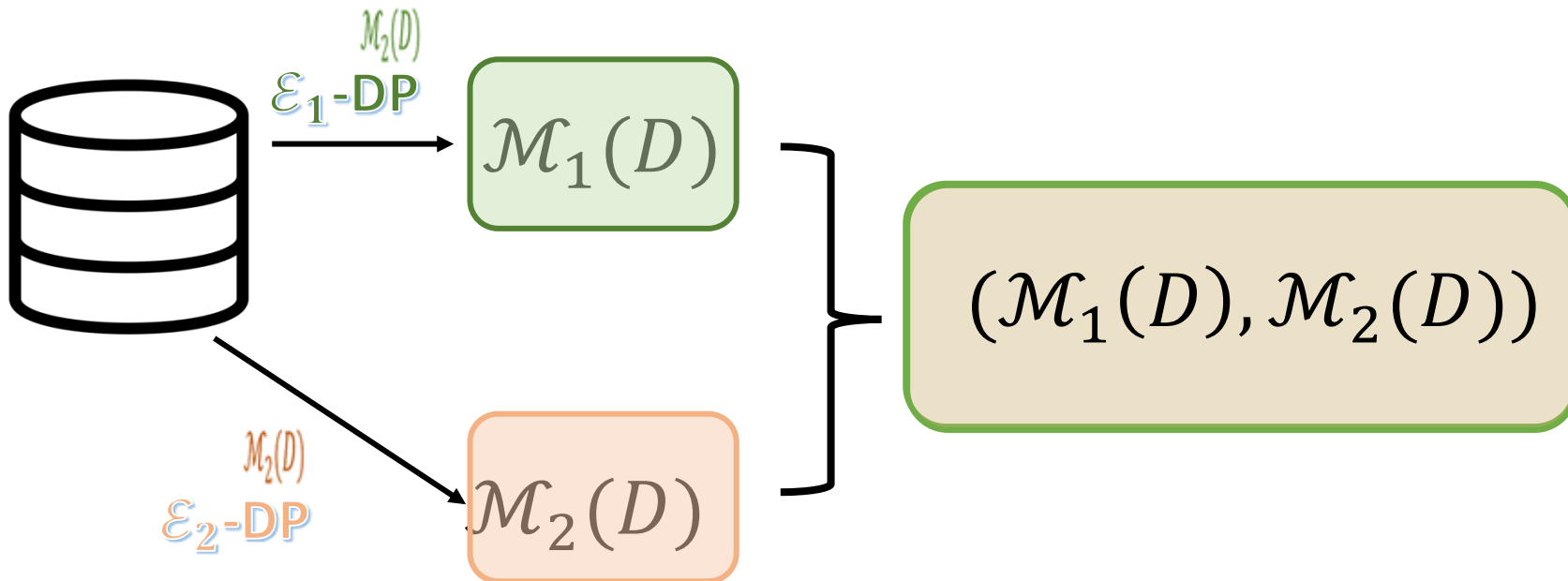


No matter how we will use the result of the DP mechanism, the final output keeps preserving ϵ –differential privacy, **AS LONG AS WE DO NOT ACCESS THE PRIVATE DATA**

Differential Privacy properties

Composition Theorem

Let $\mathcal{M}_1: \Omega \rightarrow \mathcal{R}$ and $\mathcal{M}_2: \Omega \rightarrow \mathcal{S}$ be two mechanisms preserving, respectively, ϵ_1 -differential privacy and ϵ_2 -differential privacy, with $\epsilon_1, \epsilon_2 > 0$. Then the mechanism $\mathcal{N}: \Omega \rightarrow \mathcal{R} \times \mathcal{S}$ such that $\mathcal{N}(D) = (\mathcal{M}_1(D), \mathcal{M}_2(D))$ preserves $(\epsilon_1 + \epsilon_2)$ -differential privacy.

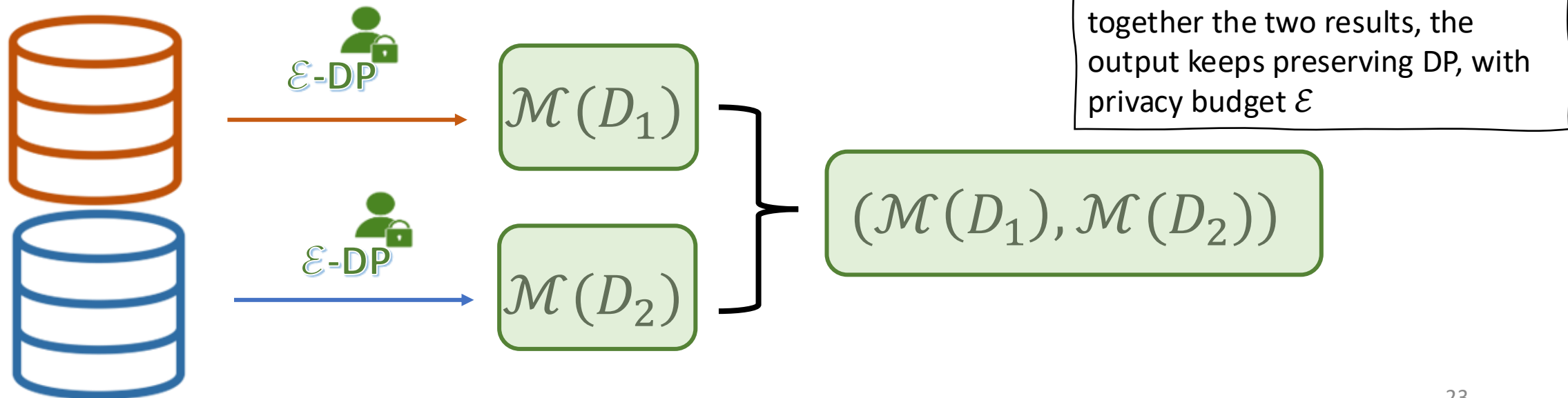


No matter how we compose together two DP outcomes, the final outcome keeps preserving DP, with privacy budget $(\epsilon_1 + \epsilon_2)$, **AS LONG AS WE DO NOT ACCESS THE PRIVATE DATA**

Differential Privacy properties

Parallel Composition Theorem

Let $D \in \Omega$ be a dataset and let $\mathcal{M}: \Omega \rightarrow \mathcal{R}$ be a mechanism preserving $\mathcal{E}$ -differential privacy. Let $D_1 \cup D_2$ be a partition of D . Mechanism $\mathcal{N}: \Omega \rightarrow \mathcal{R} \times \mathcal{R}$ such that $\mathcal{N}(D) = (\mathcal{M}(D_1), \mathcal{M}(D_2))$ preserves $\mathcal{E}$ -differential privacy



Example – Counting queries (3)

Few slides ago we have computed query $q(D) = (n, n_c)$ in a DP way, using the Laplace mechanism with privacy budget $\varepsilon = 1$. We have proved that the global sensitivity of q is 2 and thus the Laplace mechanism is

$$(\tilde{n}, \tilde{n}_c) = (n + \text{Lap}(0,2), n_c + \text{Lap}(0,2))$$

Another way to address the same problem is to consider function q as the composition of two queries $q_1(D) = n$ and $q_2(D) = n_c$. Both the queries have sensitivity 1 and we can compute separately

$$\tilde{n} = n + \text{Lap}(0,1) \quad \text{and} \quad \tilde{n}_c = n_c + \text{Lap}(0,1)$$

The composition theorem states that the overall function that releases $(\tilde{n}, \tilde{n}_c)$ preserves 2-DP ($\varepsilon = 1 + 1$).

If we want the overall function to preserve 1-DP ($\varepsilon = 1$), we must use half of the privacy budget ($\varepsilon/2$) for each one of the functions, obtaining

$$\tilde{n} = n + \text{Lap}(0,2/\varepsilon) \quad \text{and} \quad \tilde{n}_c = n_c + \text{Lap}(0,2/\varepsilon)$$

Same Result!!!

Example – Counting queries (4)

Instead of computing $q(D) = (n, n_c)$ we might prefer to compute in a differentially private way function $q'(D) = (n_o, n_c)$, where n_o is the number of students attending the class online and n_c is the number of students attending the class physically.

As usual, both the count of the students online and that of the students physically in class have global sensitivity 1. However, the two functions act on **disjoint subsets of the dataset** (a student can be either online or in class, not both). Thus, the parallel composition theorem assures that the function

$$(\tilde{n}_o, \tilde{n}_c) = (n_o + \text{Lap}(0,1), n_c + \text{Lap}(0,1))$$

preserves 1-DP.

The information of the (noised) total number of students can be computed as $\tilde{n} = \tilde{n}_o + \tilde{n}_c$. This computation does not require the injection of further noise, because its result does not depend on the original secret dataset (post-processing theorem).

(ϵ, δ) -differential privacy

Relaxed version of differential privacy definition, in which a tolerance term $\delta \in (0,1)$ has been added.

A mechanism $\mathcal{M}: \Omega \times Q \rightarrow \mathcal{R}$ preserves (ϵ, δ) -differential privacy if, for any query q , any pair of adjacent datasets $D, D' \in \Omega$ and any outcome $r \in \mathcal{R}$,

$$P(\mathcal{M}(q, D) = r) \leq e^\epsilon \cdot P(\mathcal{M}(q, D') = r) + \delta$$

Theorem:

$\mathcal{M}$ preserves (ϵ, δ) -differential privacy if there exists an event E with $P(E) > 1 - \delta$ such that $\mathcal{M}$ preserves ϵ -differential privacy when E happens.

(ϵ, δ) -differential privacy

- δ represents a **failure probability** of the mechanism.
- Composition and post-processing theorems yield also for this privacy definition: the δ parameters of the different mechanisms we want to compose must be added together (exactly as we do for ϵ).
- When $\delta=0$, the two DP definitions coincide:
$$(\epsilon, 0)\text{-DP} = \epsilon\text{-DP}$$
- (ϵ, δ) -differential privacy, with $\delta > 0$, is weaker than ϵ -differential privacy. In this case, we admit that the mechanism can fail and the privacy cannot be preserved in special cases.

(ϵ, δ) -differential privacy - Examples

$q(D)$ = number of records of D .

- **Mechanism 1.** Randomly extract a number $x \in (0,1)$. Then return

$$\mathcal{M}(D) = \begin{cases} q(D) + \text{Lap}(\frac{GS}{\epsilon}), & x \geq \delta \\ D, & x < \delta \end{cases}$$

- **Mechanism 2.** Randomly extract a number $x \in (0,1)$. Then return

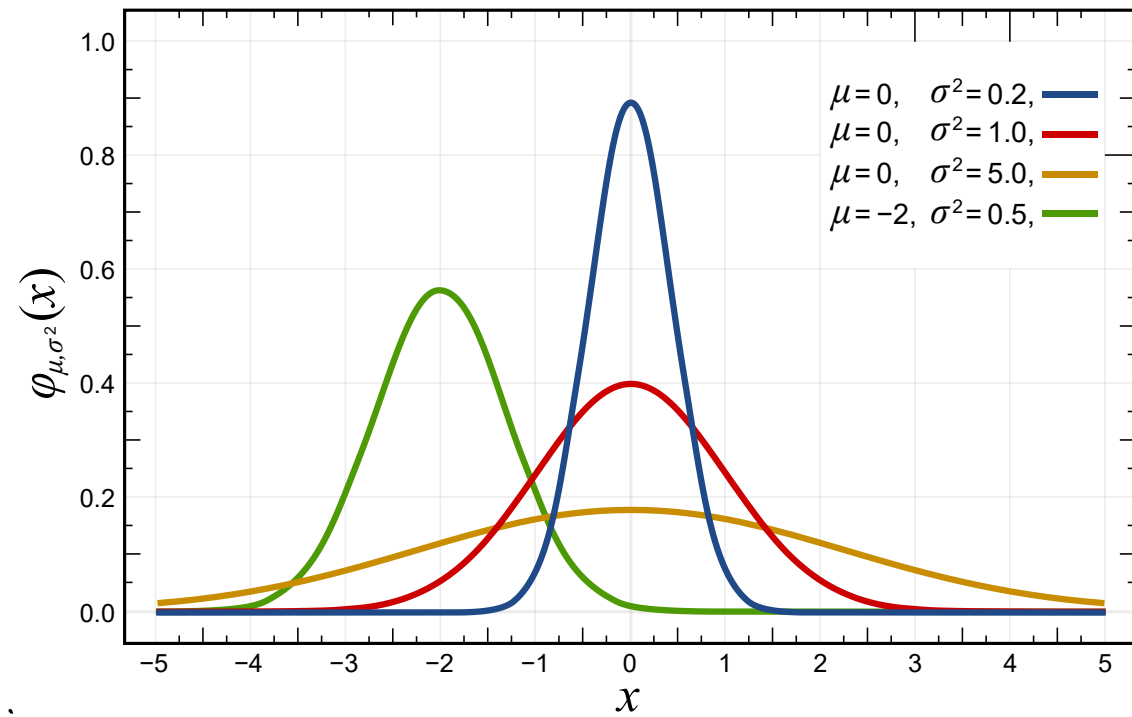
$$\mathcal{M}(D) = \begin{cases} q(D) + \text{Lap}(\frac{GS}{\epsilon}), & x \geq \delta \\ q(D), & x < \delta \end{cases}$$

Both mechanisms preserve (ϵ, δ) -differential privacy, but the first one is more dangerous, because in case of failure it releases the entire dataset:

- δ must be as small as possible
- The way in which the mechanism works is crucial in the choice of δ

Gaussian Mechanism

It is an alternative to Laplace mechanism: additive mechanism only working with numeric functions



$$\mathcal{M}_{Gauss}: (q, D) \mapsto q(D) + \mathcal{N}(0, \sigma^2)$$

Gaussian Noise ←

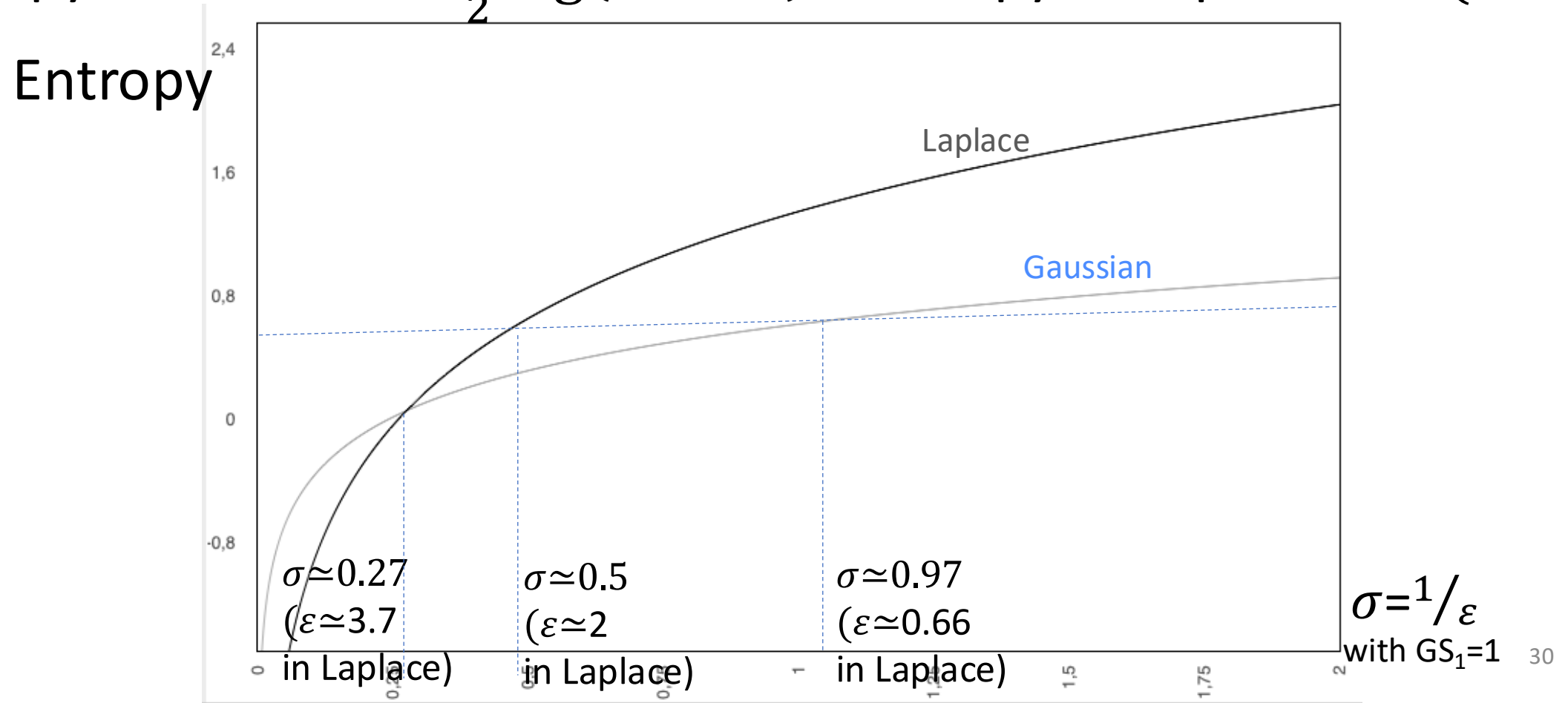
Theorem: given $\varepsilon, \delta \in (0,1)$, the Gaussian Mechanism preserves (ε, δ) -differential privacy if

$$\sigma^2 = \frac{2GS_2(q)^2 \log \frac{1.25}{\delta}}{\varepsilon^2}$$

← $\sqrt{\pi/2}$

Entropy of the Laplace and Gaussian distributions as functions of standard deviation σ

Entropy of Gaussian = $\frac{1}{2} \log(2\pi e \sigma^2)$; Entropy of Laplace = $\ln(\sqrt{2} e \sigma)$



L2 Global Sensitivity

Let $q: \Omega \rightarrow \mathbb{R}^n$ be a numeric function.

While Laplace mechanism scales the noise on the L1 global sensitivity

$$GS_1(q) = \max_{D \sim D'} \|q(D) - q(D')\|_1 = \max_{D \sim D'} \left(\sum_{k=1}^n |q_k(D) - q_k(D')| \right)$$

Gaussian mechanism looks at the L2 global sensitivity

$$GS_2(q) = \max_{D \sim D'} \|q(D) - q(D')\|_2 = \max_{D \sim D'} \sqrt{\sum_{k=1}^n (q_k(D) - q_k(D'))^2}$$

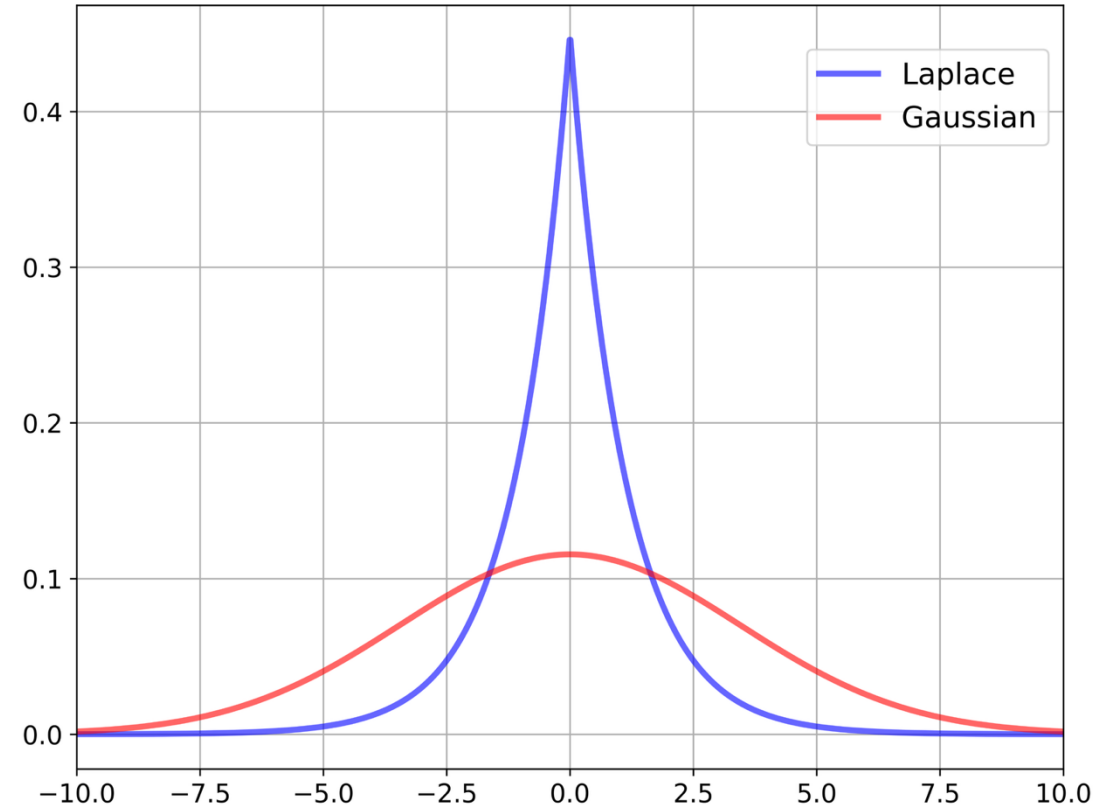
Laplace vs Gauss

LAPLACE

- ϵ -differential privacy
- It works with any $\epsilon > 0$
- It scales the noise on L1 GS
- With the same GS and ϵ , it adds less noise

GAUSS

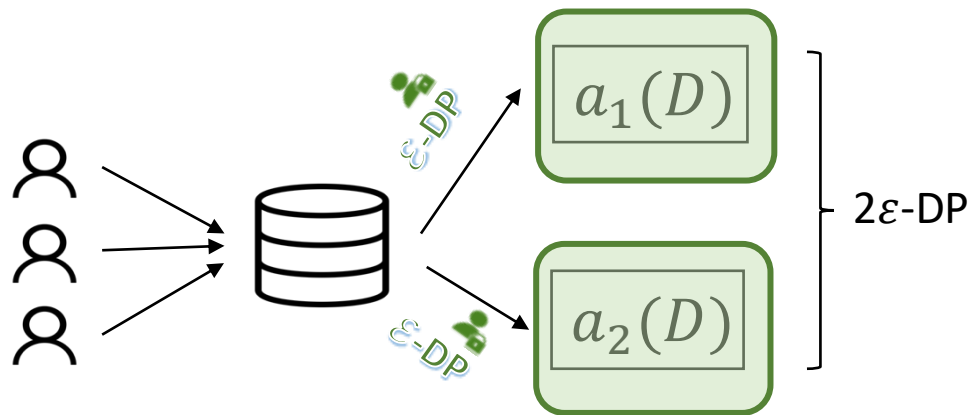
- (ϵ, δ) -differential privacy
- Privacy guaranteed only for $\epsilon < 1$
- It scales the noise on L2 GS: better mechanisms when $GS_2(q) \ll GS_1(q)$



Interactive vs. Non-interactive scenario

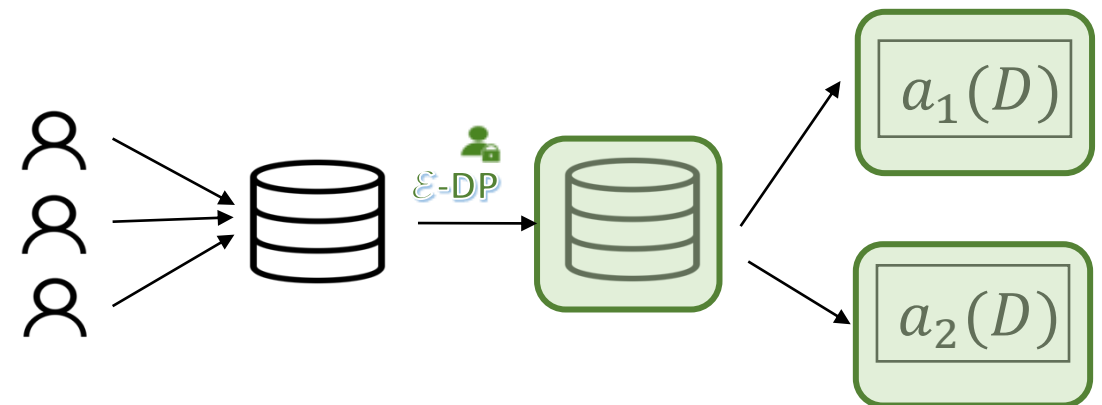
INTERACTIVE SCENARIO

- The dataset is kept secret
- Many analysis can be done on dataset D , but each one consumes part of the privacy budget



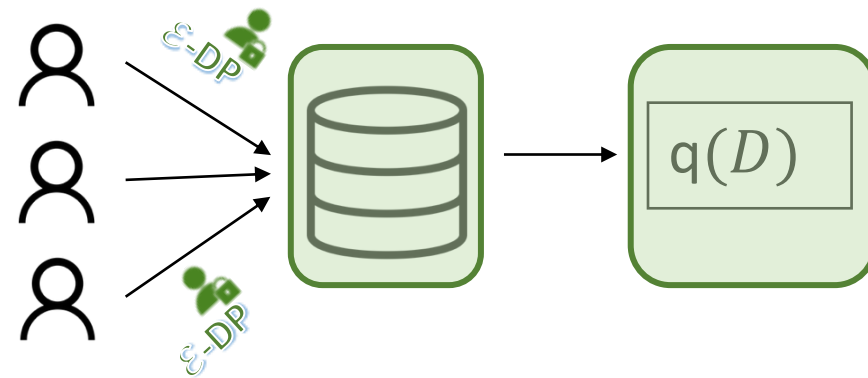
NON-INTERACTIVE SCENARIO

- The dataset is released privately:
 - Synthetic dataset somehow similar to the secret one
 - Private synopsis of the data
- Any further analysis preserves DP, with no privacy budget consumption (post-processing theorem)



Untrusted curator

- All the previous techniques assume the curator to be trusted
- What can we do if the curator is untrusted and the users do not want her or him to know their data?



- Each user injects noise on their own data and only reveals the noised version of their records

(ϵ, δ) -local differential privacy

Let $\mathcal{U}$ be the set of all possible data records.

A mechanism $\mathcal{M}$ preserves (ϵ, δ) -local differential privacy if, for any query $q: \mathcal{U} \rightarrow \mathcal{R}$, any possible **pair of records** $x, x' \in \mathcal{U}$ and any possible outcome $r \in \mathcal{R}$,

$$P(\mathcal{M}(q, x) = r) \leq e^\epsilon \cdot P(\mathcal{M}(q, x') = r) + \delta$$

(ϵ, δ) -local differential privacy

DP

$$P(\mathcal{M}(q, D) = r) \leq e^\epsilon \cdot P(\mathcal{M}(q, D') = r) + \delta$$

- $q: \Omega \rightarrow \mathcal{R}$
- Indistinguishability must hold for any pair of adjacent datasets
- Post-processing and composition theorems yield
- It usually adds less noise
- It requires a trusted curator

LDP

$$P(\mathcal{M}(q, x) = r) \leq e^\epsilon \cdot P(\mathcal{M}(q, x') = r) + \delta$$

- $q: \mathcal{U} \rightarrow \mathcal{R}$ ($\Omega = \prod_{n=1}^{\infty} \mathcal{U}^n$)
- Indistinguishability must hold for **any pair of records**
- Post-processing and composition theorems yield
- It usually adds **more** noise
- It can be used when the curator is not trusted or even in a decentralized scenario

Randomized Response



An early example of privacy by randomized process is randomized response, a technique developed in the social sciences to collect statistical information about embarrassing or illegal behavior, captured by having a property P

Study participants are told to report whether or not they have property P as follows:

1. Flip a coin.
2. If tails, then respond truthfully.
3. If heads, then flip a second coin and respond “Yes” if heads and “No” if tails

The Randomized Response preserves $\log 3$ -differential privacy

Proof. $\mathcal{U} = \{\text{all possible study participants}\}$. $q: \mathcal{U} \rightarrow \{\text{Yes}, \text{No}\}$. $\forall x \in \mathcal{U}$, $\mathcal{M}(q, x)$ is the output of the randomized response.

Let x be a study participant such that $q(x) = \text{Yes}$.

- $P(\mathcal{M}(q, x) = \text{Yes}) = P(\mathcal{M}(q, x) = \text{Yes} \mid \text{Tails}) \cdot P(\text{Tails}) + P(\mathcal{M}(q, x) = \text{Yes} \mid \text{Heads}) \cdot P(\text{Heads}) = \frac{1}{2} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{3}{4}$
- $P(\mathcal{M}(q, x) = \text{No}) = P(\mathcal{M}(q, x) = \text{No} \mid \text{Tails}) \cdot P(\text{Tails}) + P(\mathcal{M}(q, x) = \text{No} \mid \text{Heads}) \cdot P(\text{Heads}) = \frac{1}{2} \cdot \frac{1}{2} + 0 \cdot \frac{1}{2} = \frac{1}{4}$

Similarly, if x is such that $q(x) = \text{No}$, $P(\mathcal{M}(q, x) = \text{Yes}) = \frac{1}{4}$ and $P(\mathcal{M}(q, x) = \text{No}) = \frac{3}{4}$.

Let x, y be two participants. Of course, if $q(x) = q(y)$ then $P(\mathcal{M}(q, x) = r) = P(\mathcal{M}(q, y) = r)$, for each $r \in \{\text{Yes}, \text{No}\}$ and indistinguishability between x and y is guaranteed for any ϵ .

Consider x, y s.t. $q(x) = \text{Yes}$ and $q(y) = \text{No}$.

$$\frac{P(\mathcal{M}(q, x) = \text{Yes})}{P(\mathcal{M}(q, y) = \text{Yes})} = \frac{3/4}{1/4} = 3 = e^{\log 3}$$

$$\frac{P(\mathcal{M}(q, x) = \text{No})}{P(\mathcal{M}(q, y) = \text{No})} = \frac{1/4}{3/4} = \frac{1}{3} < 3 = e^{\log 3}$$

$$\frac{P(\mathcal{M}(q, y) = \text{No})}{P(\mathcal{M}(q, x) = \text{No})} = \frac{3/4}{1/4} = 3 = e^{\log 3}$$

$$\frac{P(\mathcal{M}(q, y) = \text{Yes})}{P(\mathcal{M}(q, x) = \text{Yes})} = \frac{1/4}{3/4} = \frac{1}{3} < 3 = e^{\log 3}$$

Generalized Randomized Response

$q: \mathcal{U} \rightarrow \mathcal{R}$, where $\mathcal{R}$ is a discrete set with cardinality $|\mathcal{R}| = k$.

The generalized randomized response is the mechanism $\mathcal{M}$ which returns a value $v \in \mathcal{R}$ with probability

$$P(\mathcal{M}(q, x) = v) = \begin{cases} \frac{e^\varepsilon}{e^\varepsilon + k - 1} & \text{if } q(x) = v \\ \frac{1}{e^\varepsilon + k - 1} & \text{otherwise} \end{cases}$$

It can be proved that the generalized randomized response preserves ε -differential privacy.

Methods to preserve LDP

- $q(x)$ returns a categorical value: Generalized Randomized Response
- $q(x)$ returns a numerical value: Laplace/ Gaussian Mechanism

$$q(x) + \text{Lap}(0, \frac{GS_1(q)}{\epsilon})$$

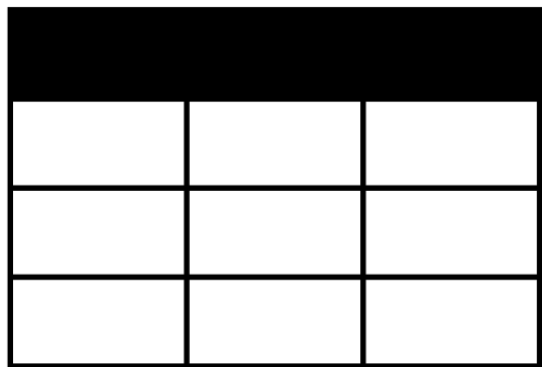
but this time $GS(q)$ is the maximum variation when q is computed on two different records x, x' .

$$GS_1(q) = \max_{x, x' \in \mathcal{U}} ||q(x) - q(x')||_1$$

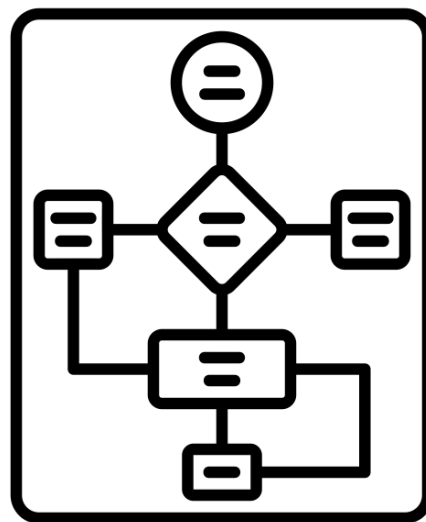
Machine Learning and Differential Privacy

DP Machine Learning

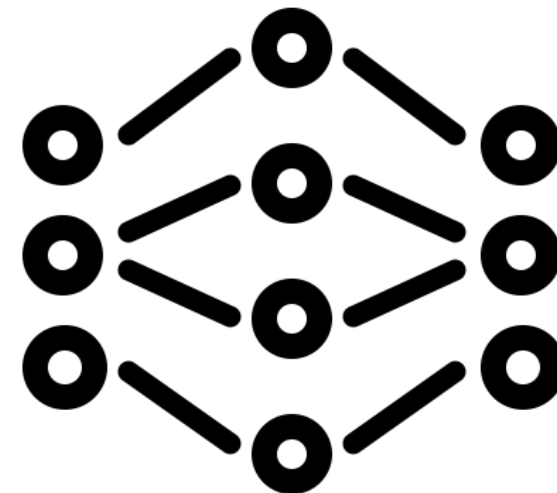
- In the last years, many differentially private machine learning algorithms have been developed
- The idea of DP Machine Learning is to inject some noise in the training phase of the ML model, in order to hide the presence of a particular record in the training set
- It is important that a machine learning model does not depend too heavily on the training data (to avoid overfitting). In this sense, ML and DP have a common goal: to return accurate results which do not depend on the presence of a particular individual in the data.



Training Set



Algorithm



Model

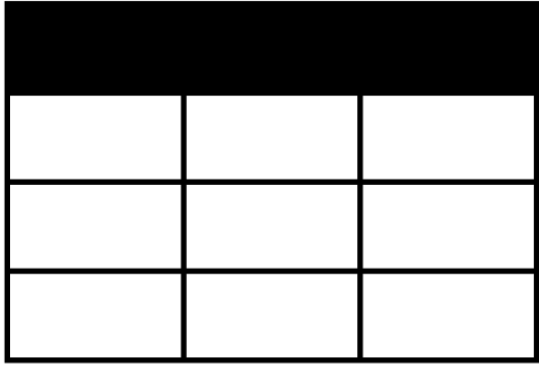
$$f_{\theta}: x \mapsto \{C_1, C_2, \dots, C_r\}$$

Among all the functions of a fixed kind, the one that better describes the data is chosen.

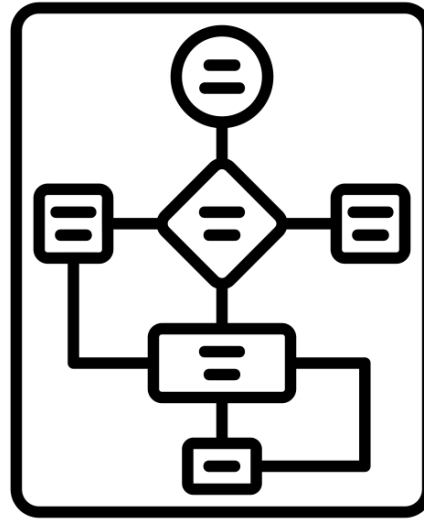
Usually, training a model means to find the best value of a parameter θ that:

$$A: \Omega \longrightarrow \mathbb{R}^d$$

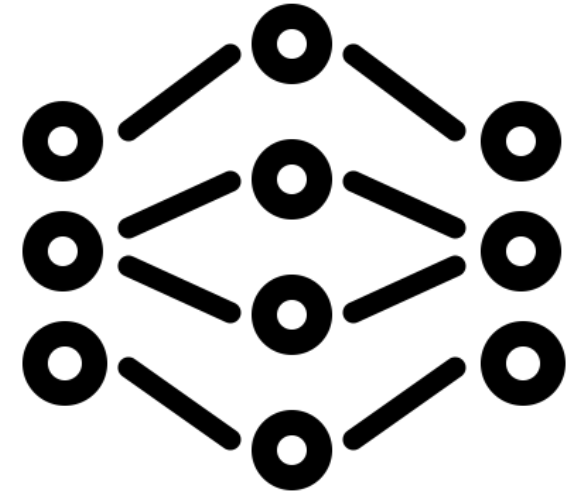
$$D \longmapsto \theta$$



Training Set



Algorithm



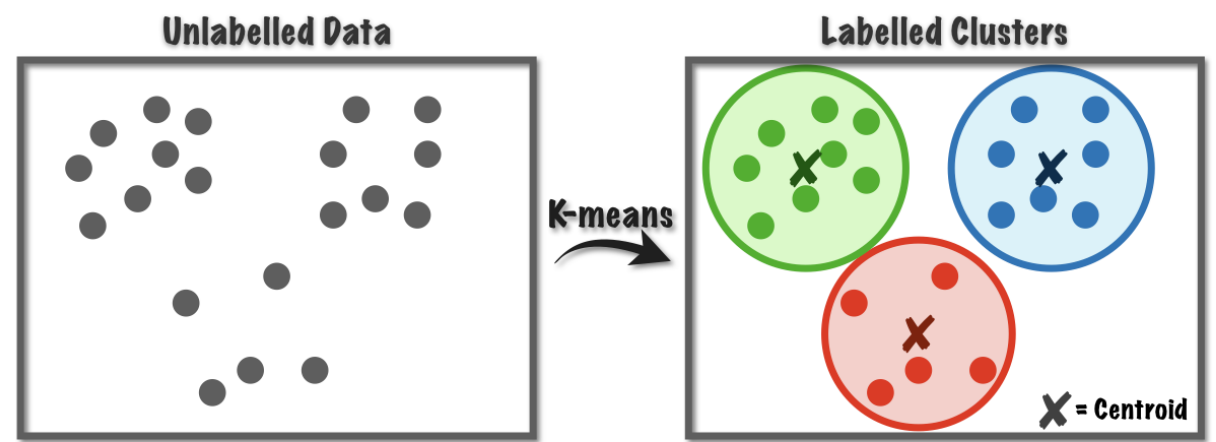
Model

Input
Perturbation

In-process
Mechanisms

Output
Perturbation

DP-k-means

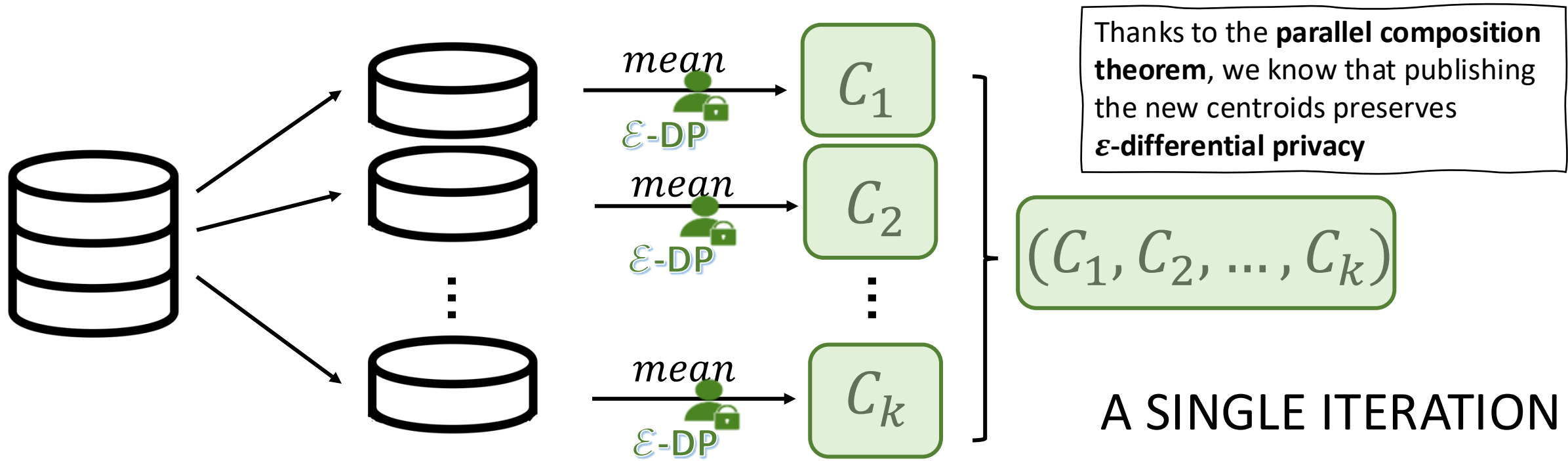


$$C_1, \dots, C_k \text{ minimizing } \sum_{i=1}^k \sum_{x \in C_i} d(x, C_i)$$

The most popular algorithm to identify the clusters' centroids is Lloyd's algorithm.

At each iteration, given the initial data and the centroids computed at the previous iteration:

1. Each data point is assigned to the cluster corresponding to the nearest centroid (of the previous iteration)
2. The new centroids are computed as mean of the coordinates of the points in each cluster using the Laplace mechanism



Each iteration preserves **ϵ -differential privacy**. If n iterations are performed, the overall algorithm will preserve **$(n \cdot \epsilon)$ -differential privacy** (composition theorem).

There is a last detail to understand: how can we compute the mean in a DP way?

How to compute the mean of the coordinates of a cluster in a DP-way

Let C be a cluster. Each $x \in C$ is a point in $\mathbb{R}^d$, i.e. $x = (x_1, \dots, x_d)$. Also the mean of all the points in C will be a d -dimensional point

where $m_i = \frac{\sum_{x \in C} x_i}{|C|}$, for each $i=1, \dots, d$.

To obtain m we can use the Laplace mechanism to privately compute

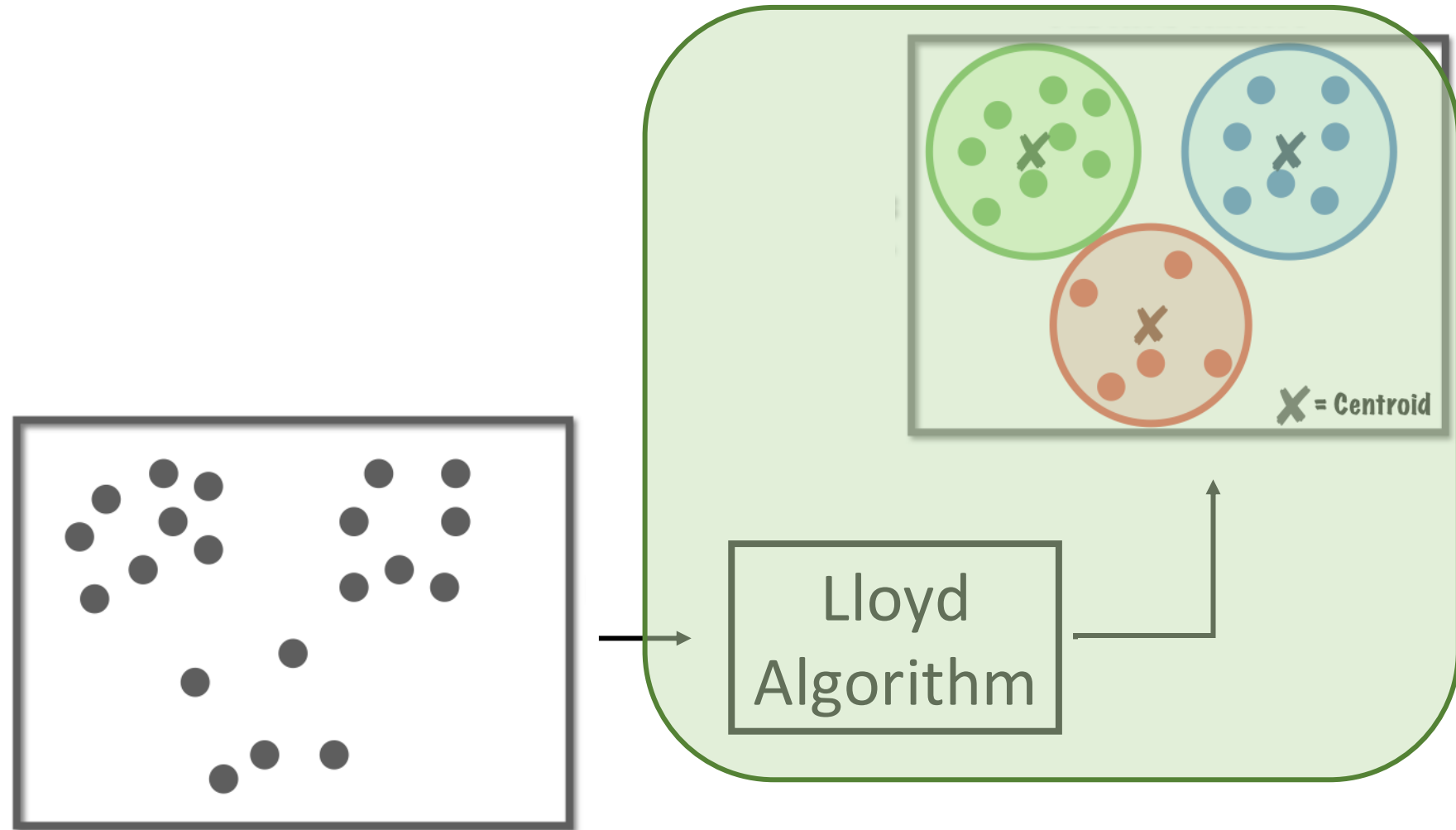
$$q(C) = (|C|, \sum_{x \in C} x_1, \dots, \sum_{x \in C} x_d)$$

When adding one point to C , the count $|C|$ varies of 1. Instead the sum of the i -th coordinates varies of the i -th coordinate of the new point. Let us suppose that the points are all belonging to the hypersquare $[-1, 1]^d$: then the maximum variation is 1. Putting all together, the global sensitivity of q is

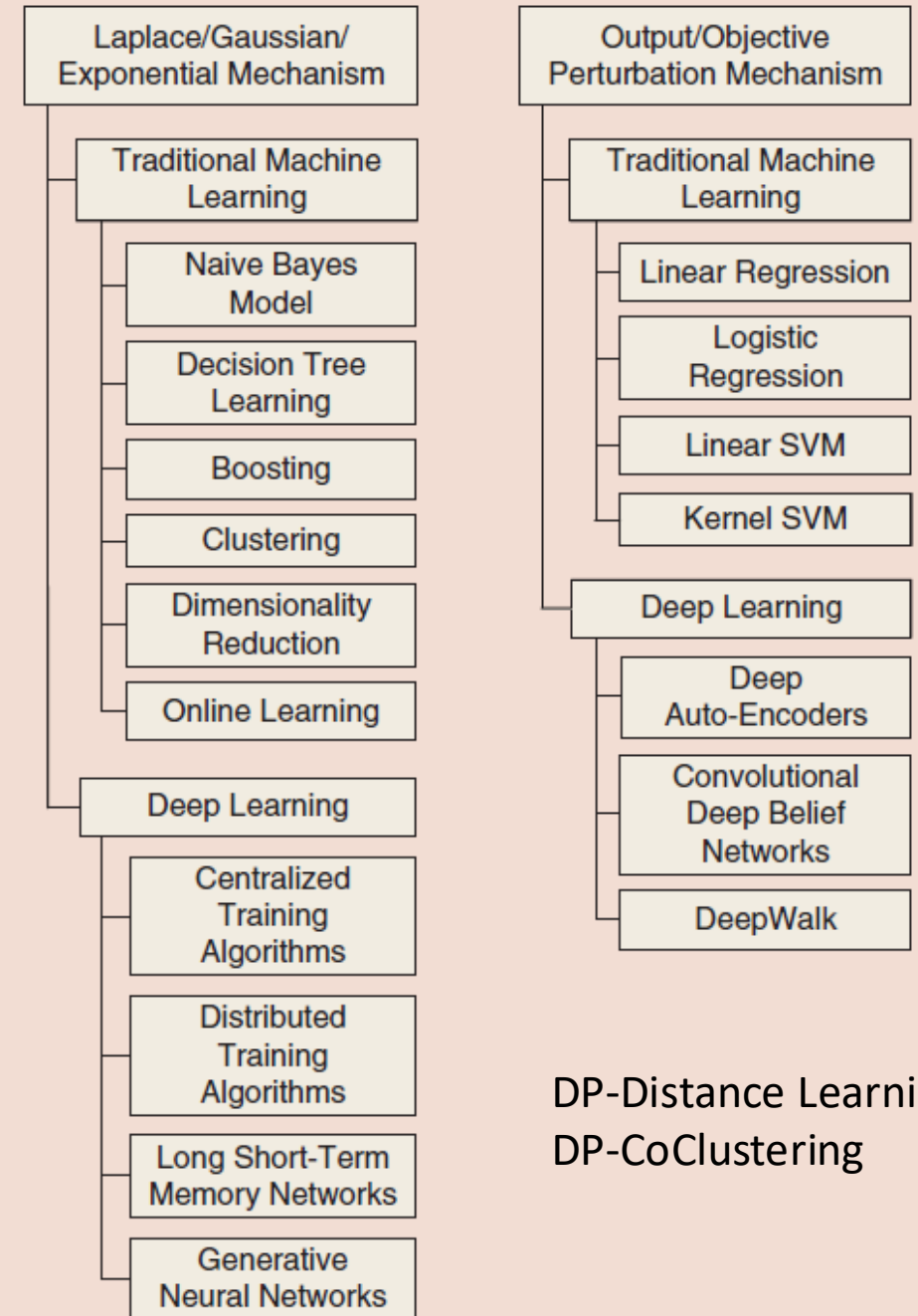
$$GS(q) = 1 + d$$

Finally, we can compute $q(C)$ using the Laplace mechanism with noise parameter $\frac{1+d}{\epsilon}$ and then computing the mean of all the coordinates.

DP-k-means (input perturbation)



Other results in DP-Machine Learning



DP-Distance Learning
DP-CoClustering

Limitations and Critical Issues

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Critical issues

- How to choose ϵ ?
- What can we do when the function has high global sensitivity?
- Controlling the number of iterations: for iterative algorithms, usually each iteration is computed in a differentially private way. The overall algorithm preserves $n\epsilon$ -DP (composition theorem), where n is the total number of iterations. Hence, the greater n , the lower the privacy level.
- Convergence issues: let us suppose to have an iterative algorithm $\mathcal{A}$ which converges to a solution. When the noise is added to each iteration in order to preserve DP, usually the convergence of $\mathcal{A}$ is no longer guaranteed.
- Results out of domain: let $q: \Omega \rightarrow [0,1]$ and suppose to use the Laplace mechanism to compute $q(D)$. The Laplace mechanism can return any possible real value; for instance, it returns $\widetilde{q(D)} = 5$ which is outside the domain of q . Possible solutions are: repeating the mechanism until it returns a valid mechanism; project the result in the domain (post-processing theorem guarantees that the result is still DP).
- Setting the parameters/hyperparameters when the data are not accessible

How to choose ϵ ?

- When ϵ is small (close to 0), the meaning of the definition is clear.
- Let us set $\epsilon = 8$. DP definition requires that

$$\frac{P(\mathcal{M}(D,q)=r)}{P(\mathcal{M}(D',q)=r)} \leq e^8 \approx 2981$$

- For instance, it is allowed that

$$P(\mathcal{M}(D, q) = r) = 1 \text{ and } P(\mathcal{M}(D', q) = r) = \frac{1}{2981} = 0.0003$$

- Although the definition of differential privacy is perfectly satisfied, it is easy to understand, from the result r , that the secret dataset is D .

How to choose ϵ ?

“Using differential privacy to maximize learning while providing a meaningful degree of privacy requires judicious choice of the parameter ϵ (among other factors). But there is little understanding of what is a judicious value of ϵ for a given system [...]

We do not know what parameter ϵ is right for any given differentially private analysis, and we do know that the answer can vary tremendously based on attributes of the dataset and the policies and practices that constrain those who query it.”

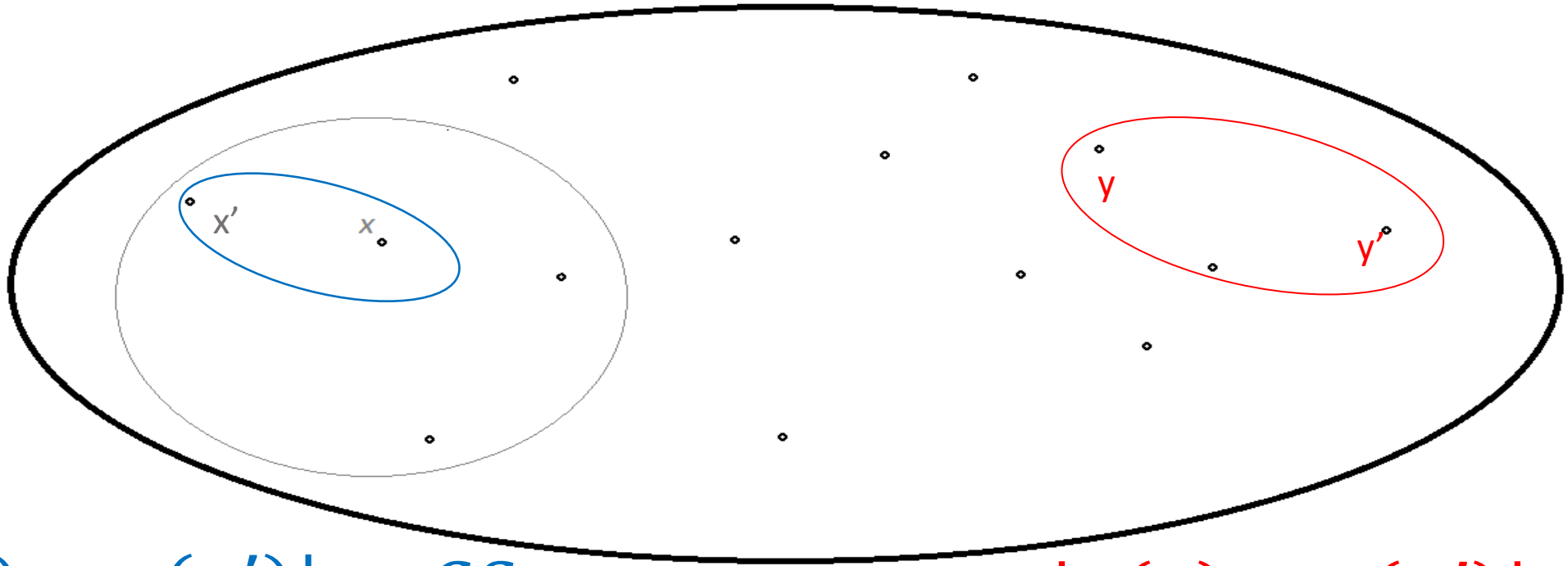
C. Dwork

Differential Privacy in Practice: Expose your Epsilon (2019)

Issues related to the Global Sensitivity

- The global sensitivity of a function can be high
- Some functions even have infinite global sensitivity
- Sometimes, it is complicated to analytically derive the global sensitivity of a function
- POSSIBLE SOLUTIONS:
 - Using the local sensitivity of the function;
 - Estimate the global sensitivity of the function from data;
 - There exist few methods that are not based on the sensitivity.

Local sensitivity



$$|q(x) - q(x')| < GS_q$$

$$|q(y) - q(y')| = GS_q$$

Local sensitivity

Remind that the global sensitivity of a query q is

$$GS_1(q) = \max_{(D, D') \text{ s.t. } D \sim D'} ||q(D) - q(D')||_1$$

Instead, the local sensitivity of q in D is

$$LS_1(q, D) = \max_{D' \text{ s.t. } D \sim D'} ||q(D) - q(D')||_1$$

Unfortunately, the Laplace mechanism calibrated on the local sensitivity does not preserve DP

$$LS_1(q, D) \leq GS_1(q), \quad \forall D \in \Omega$$

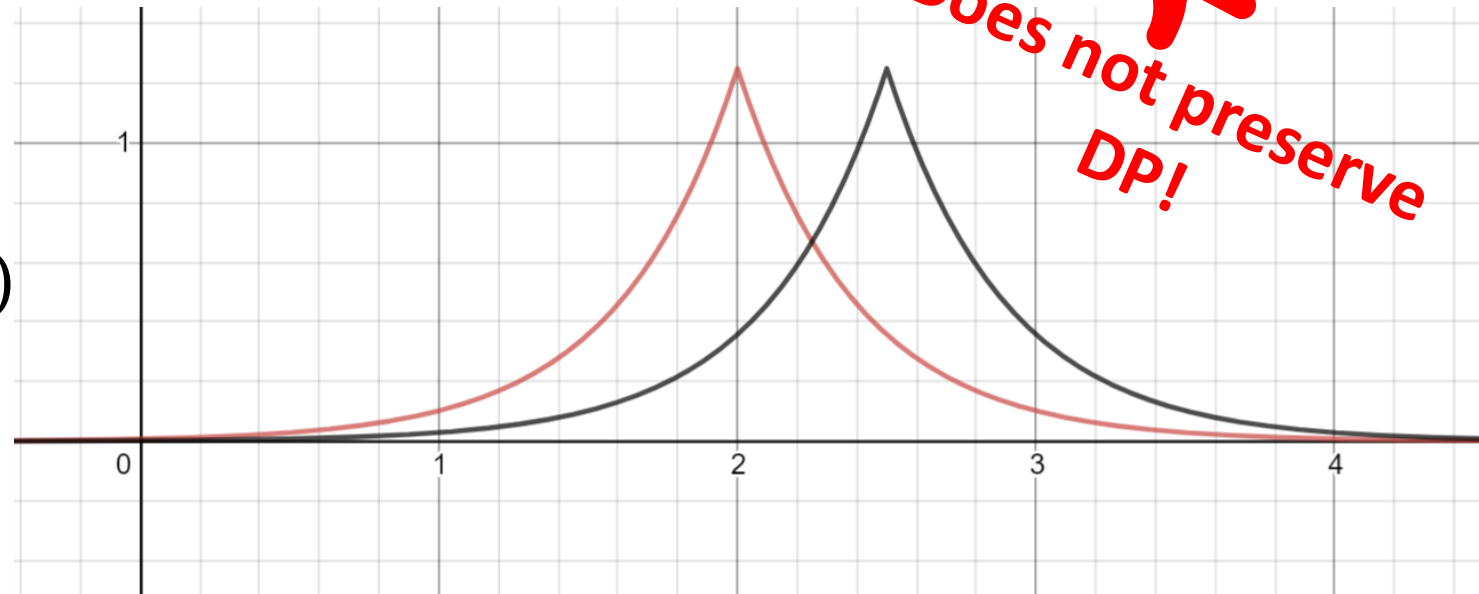
Using the LS to calibrate the noise instead of GS increases the accuracy of the mechanism!

Laplace mechanism with LS (first attempt)

First Method: using the Laplace mechanism with scale parameter fixed to $\sigma = LS_1(q, D)/\epsilon$, i.e. the shape of the Laplacian is the same whatever the secret dataset is.

Thus:

- $\mathcal{M}(D, q) = Lap(q(D), \sigma)$
- $\mathcal{M}(D', q) = Lap(q(D'), \sigma)$

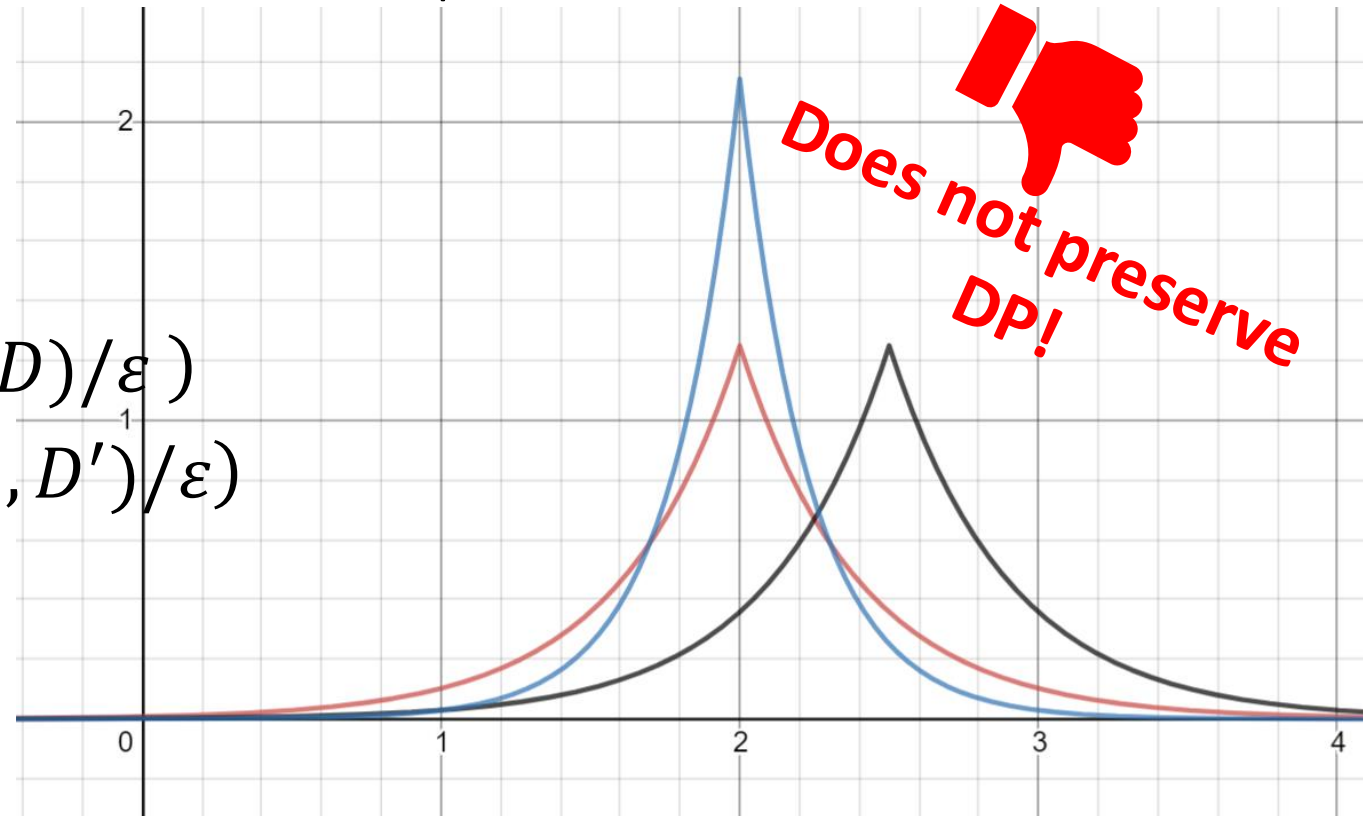


Laplace Mechanism with LS (second attempt)

Second Method: using the Laplace mechanism with scale parameter $LS_1(q, \cdot)/\epsilon$, i.e. the shape of the noise curve depends on the secret dataset.

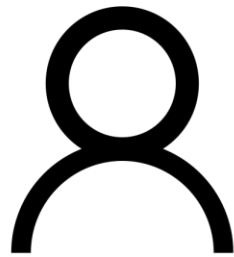
Thus,

- $\mathcal{M}(D, q) = \text{Lap}(q(D), LS_1(q, D)/\epsilon)$
- $\mathcal{M}(D', q) = \text{Lap}(q(D'), LS_1(q, D')/\epsilon)$



Propose-Test-Release

- Query q
- $GS(q)$ known to both actors
- $GS(q)$ is too high and the analyst does not trust a result with noise scaled on it



Analyst

1. Propose a threshold $b < GS(q)$



3. If the test is passed, release $q(D) + Lap(\frac{b}{\epsilon})$

Otherwise, return the error message
«I can't tell you the result»

Actually, this formulation of the test does not preserve DP: it can happen that D and D' are adjacent datasets but D passes the test and D' does not.

2. Test the proposed value b



Curator

IDEA OF THE TEST:

If $LS(q, D) < b$:

the test is passed

else:

the test is not passed

Propose-Test-Release

1. **A** proposes a threshold b
2. **C** computes the distance d between D and the nearest dataset D' having $LS(q, D') > b$
3. The Laplace mechanism is used to compute $\hat{d} = d + \text{Lap}(\frac{1}{\epsilon})$
4. If $\hat{d} > \ln(1/2\delta) / \epsilon$, the test is passed; otherwise, a message error is returned.



Preserves (ϵ, δ) -DP!

Errors

- To correctly apply DP is not a trivial task. There are many possible sources of error. As a result, the algorithm which is declared to be private does not actually preserve differential privacy.
 - Errors in the computation of the sensitivity
 - Sometimes the secret data are accessed without injecting noise
 - Misuse of the composition theorems
 - Different definitions of adjacent datasets
 -
- Errors are so frequent that several statistical tools for the automatic verification of the privacy guarantees of an algorithm have been developed: **StatDP**, **DpFinder**, **CheckDP**, **DPCheck**, **DiPC**

DP in the real world

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Example of application of DP



- 2020 US Census
- Uber
- Apple
- Google Chrome
- Microsoft Windows

Libraries



- google/differential-privacy (C++)
- IBM/differential-privacy-library (Python)
- brubinstein/diffpriv (R)
- tensorflow/privacy
- OpenDP (Harvard + Microsoft)

Alternative privacy definitions



- Random differential privacy
- Probabilistic differential privacy
- KL differential privacy
- Rényi differential privacy
- Zero-Concentrated differential privacy

Bibliography



- Books on DP

- Dwork, Roth, “The algorithmic foundation of Differential Privacy”, Foundations and Trends in Theoretical Computer Science, 2014
- Vadhan, “The complexity of Differential Privacy”, Tutorials on the Foundation of Cryptography, 2017

- Surveys on DP-Machine Learning

- Gong, Xie, Pan, Feng, Qin, “A Survey on Differentially Private Machine Learning”, IEEE Computational Intelligence Magazine, 2020
- Ji, Lipton, Elkan, “Differential Privacy and Machine Learning: a survey and review”, arXiv preprint arXiv:1412.7584, 2014

- Problems and Limitations

- Domingo-Ferrer, Sanchez, Blanco-Justicia, “The Limits of Differential Privacy (and its Misuse in Data Release and Machine Learning)”, Communications of the ACM, 2021
- Bambauer, Muralidhar, Sarathy, “Fool’s Gold: An Illustrated Critique of Differential Privacy”, Vanderbilt Journal of Entertainment & Technology Law, 2014

- How to choose ϵ

- Dwork, Kohli, Mulligan, “Differential Privacy in Practice: Expose your Epsilon!”, Journal of Privacy and Confidentiality, 2019
- Kohli, Laskowski, “Epsilon Voting: Mechanism Design for Parameter Selection in Differential Privacy”, IEEE Symposium on Privacy-Aware Computing, 2018
- Hsu et al, “Differential Privacy: An Economic Method for Choosing Epsilon”, IEEE Computer Security Foundations Symposium, 2014